

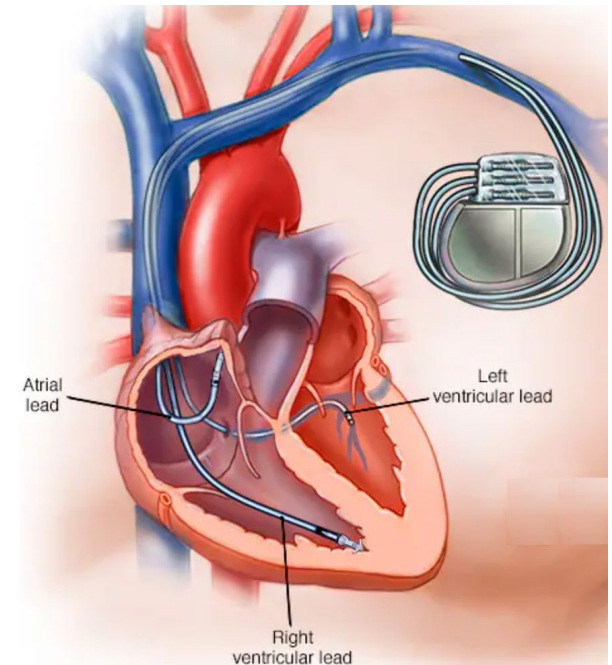
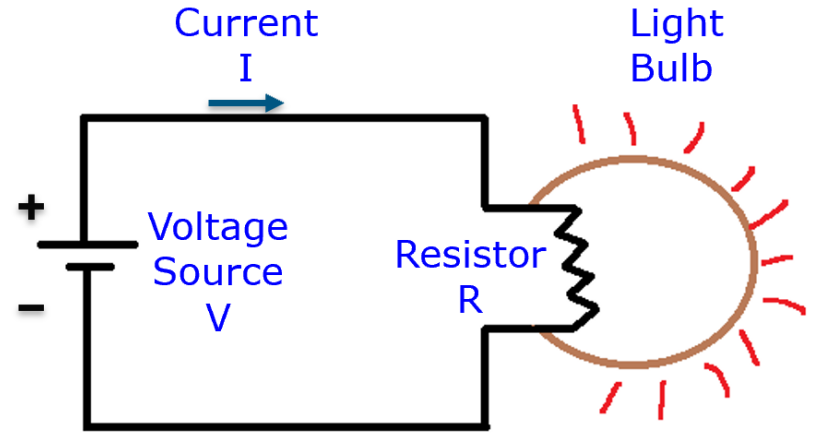
Electrical Science

ENGR 2303

Dr. Mohamed Bingabr
School of Engineering
University of Central Oklahoma

Basic Concepts of Electric Circuit

- Ohm's Law
- Electric Charge
- Current
- Resistor
- Voltage
- Energy
- Power
- Circuit Elements

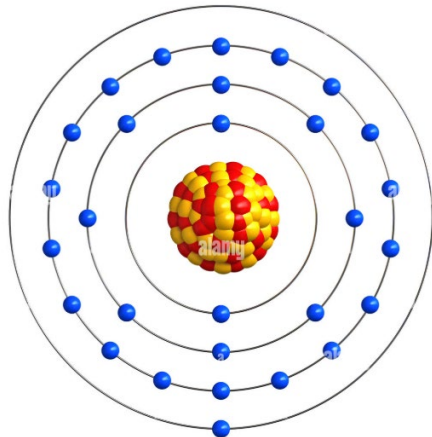


Circuit Theory Video 1

Electric Charges

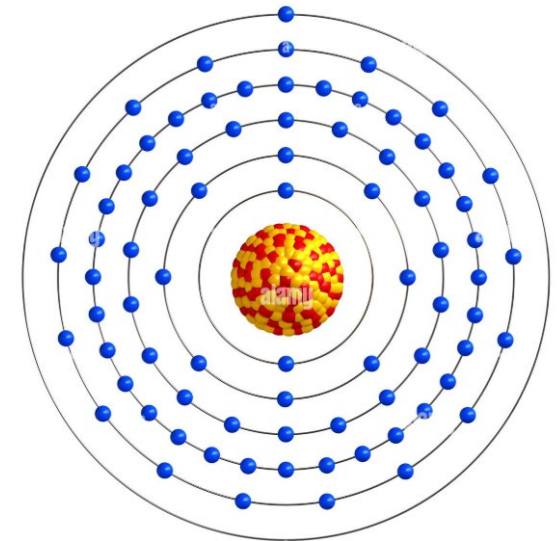
- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge **e** on one electron is negative and equal in magnitude to $1.602 \times 10^{-19} \text{ C}$ which is called as electronic charge. The charges that occur in nature are **integral multiples** of the electronic charge.

Copper



● 29Protons ● 35Neutrons ● 29Electrons

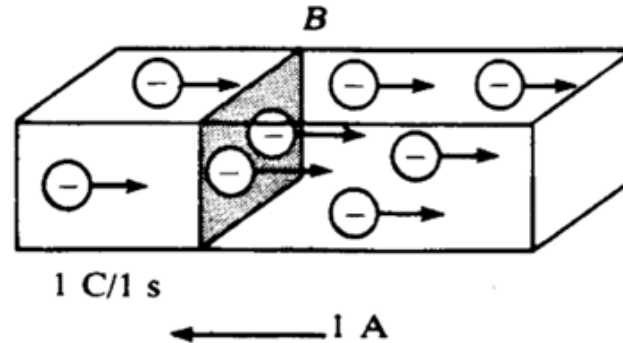
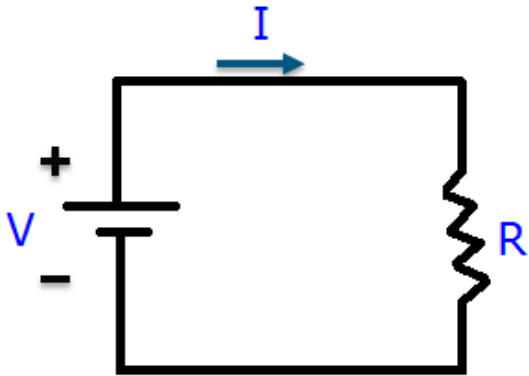
Plutonium



● 78Protons ● 117Neutrons ● 78Electrons

Current

- Electric current $i = dq/dt$. The unit of ampere can be derived as $1 \text{ A} = 1 \text{ C/s}$.



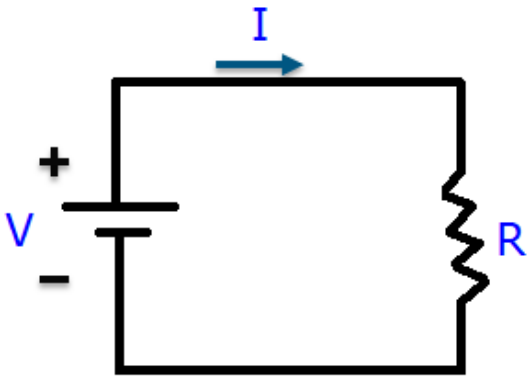
- A **direct current (dc)** is a current that remains constant with time.
- An **alternating current (ac)** is a current that varies sinusoidally with time.
(reverse direction)

A conductor has a constant current of 5 A.
How many electrons pass a fixed point on the
conductor in one minute?

Resistor

- Resistance of a material is its natural opposition to the flow of electric current. It determines how difficult it is for electrons to move through a conductor.

$$R = \frac{\rho L}{A} \quad \Omega \text{ Ohm's}$$

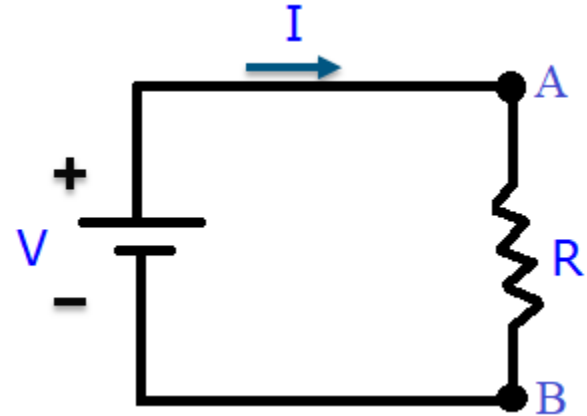


Voltage

- Voltage (or potential difference) is the **energy** required to move a **unit charge** through an element, measured in volts (V).

$$v_{ab} = dw / dq \quad (\text{volt})$$

- **w** is energy in joules (J)
- **q** is charge in coulomb (C).



- Electric voltage, v_{ab} , is always **across the circuit element** or **between two points in a circuit**.
 - $v_{ab} > 0$ means the potential of **a** is higher than potential of **b**.
 - $v_{ab} < 0$ means the potential of **a** is lower than potential of **b**.

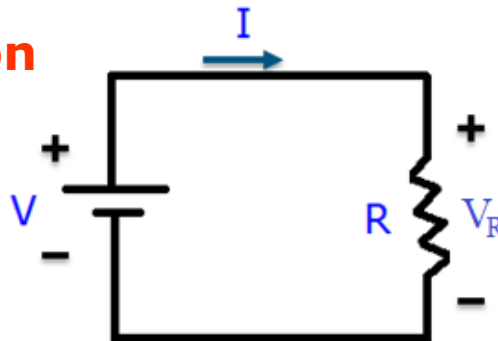
Power and Energy

- Power is the time rate of expending or absorbing energy, measured in watts (W).

- Mathematical expression: $p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$

Passive sign convention

**$p = -vi$
supplying power**



**$P = +vi$
absorbing power**

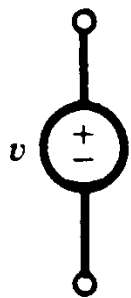
- Energy** is the capacity to do work, measured in joules (J).

- Mathematical expression $w = \int_{t_0}^t p dt = \int_{t_0}^t vi dt$

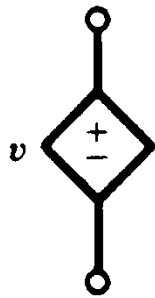
Circuit Elements

Active Elements

Passive Elements



(a)



(b)



(c)



(d)



(e)



(f)



(g)

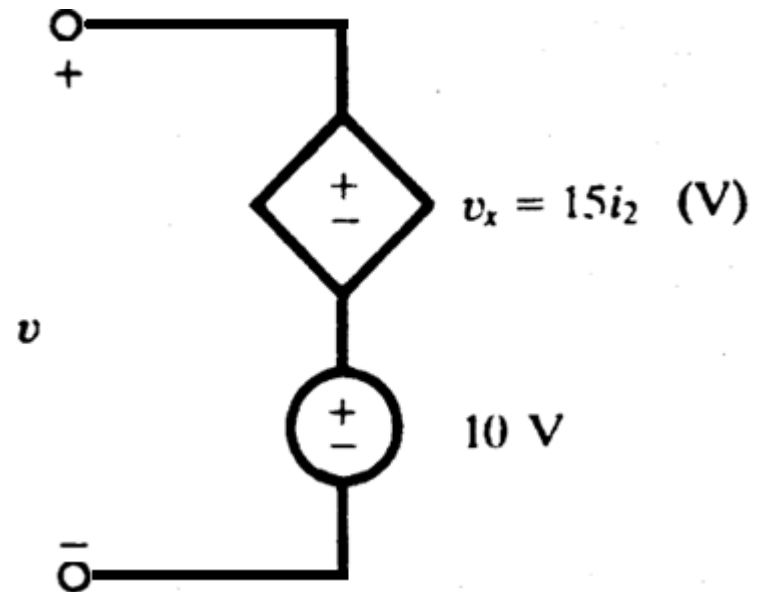
Independent sources Dependent sources

- A dependent source is an active element in which the source quantity is controlled by another voltage or current.
- They have four different types: VCVS, CCVS, VCCS, CCCS. Keep in mind the signs of dependent sources.

CCVS=Current-Controlled Voltage Source

Circuit Elements

Obtain the voltage v in the branch shown below for $i_2 = 1\text{A}$.



$$v = v_x + 10 = 15(1) + 10 = 25 \text{ V}$$